

IE684: Operations Research Lab

Metaheuristic Optimization for Routing Networks

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1. Exact Formulation vs. Metaheuristic Approximation

The Travelling Salesman Problem (TSP) seeks the optimal Hamiltonian cycle minimizing total traversal distance across a spatial network. The Dantzig-Fulkerson-Johnson (DFJ) integer formulation mathematically guarantees exact solutions by enforcing degree constraints ($\sum_{e \in \delta(i)} x_e = 2$) and dynamic Sub-tour Elimination Constraints (SECs). However, evaluating the extreme points of the TSP polytope ($\frac{(n-1)!}{2}$) becomes computationally intractable as network dimensionality scales. For a 48-node city network, evaluating $\approx 10^{59}$ combinations requires the implementation of probabilistic metaheuristics.

2. Simulated Annealing Framework

To approximate global optima on large-scale networks, a Simulated Annealing (SA) algorithm was engineered. The acceptance probability of transitioning to a geometrically swapped neighborhood state is governed by the Boltzmann distribution:

$$P = e^{-(c_{new} - c_{old})/T_k}$$

The algorithm balances state exploration against greedy exploitation via a strictly defined cooling schedule ($T_{k+1} = \alpha T_k$).

3. Benchmarking and Dimensionality Analysis

The SA engine was benchmarked across an 11-node and a 48-node routing network.

- **11-Node Network:** Demonstrated rapid convergence within 1000 iterations. Tuning the initial temperature parameter revealed that a high T_0 (1000) allowed sufficient exploration to consistently identify the global minimum, whereas a low T_0 (1) triggered premature exploitation, trapping the solver in sub-optimal local minima.
- **48-Node Network (The Curse of Dimensionality):** The exponential expansion of the state space required recalibrating the geometric cooling constant ($\alpha = 0.99$) and maximizing iterations (10,000) to artificially extend the exploration horizon. While the SA engine successfully converged toward highly efficient routing paths, the extreme dimensionality introduced inherent variance across validation runs, highlighting the heuristic nature of the approximation.